

DNA Gel Electrophoresis

Overview

Resolve specific DNA fragments to allow visual verification and confirmation of product band sizes.

Making agarose gel

A. Materials

- a. Agarose powder
- b. 1X TAE buffer
- c. Plastic wrap
- d. Agarose gel mold, tray, comb
- e. 6X DNA Safety dye (BioGreen™ Safe DNA Gel buffer)
- f. Pipette

B. Procedure

1. Assemble the tray, comb, and mold.
2. Add the appropriate amount of agarose powder according to the desired agarose concentration.
3. Add 20 mL of 1X TAE buffer to the flask and mix thoroughly.
4. Cover with plastic wrap and use a 1000µl tip to poke 4-5 holes in the plastic wrap.
5. Microwave for 30 seconds then carefully swirl or agitate the flask.
6. Repeat Step 5 until the agarose is completely dissolved and the solution becomes clear.
7. Remove the plastic wrap and wait until it is no longer smoking
8. Dilute the safe stain at a 1:20,000 ratio into the mixture and pipette
9. Pour the mixture into the mold.
10. Use the pointed end of a 1000 µL pipette tip to remove surface air bubbles, and use the flat end to gently release any air bubbles trapped at the bottom of the tray.
11. Solidify at room temperature.

Loading samples

C. Procedure

Samples and Ladder preparation

- PCR product: No dye is needed
- Miniprep product: Add 1µl of dye to every 5µl of DNA on parafilm

1. Pour 1X TAE buffer until it covers the gel
2. Place the gel in with the gel tank.
3. Add 5 μ l of DNA ladder or samples in each well
Ladder (1kb or 3kb, ladder size must exceed sample base pairs)
4. Run the gel at 100V for 30 minutes

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